Section 8.1 Dot Products and Orthogonal Sets

Back In Section 2.1 we were introduced to the dot product

The dot product: let $\vec{u} \in \mathrm{R}^{\mathbf{n}}$ $\vec{v} \in R^n$ then

$$\vec{u} \cdot \vec{v} = \begin{bmatrix} u_1 \\ u_2 \\ u_3 \\ \vdots \\ u_n \end{bmatrix} \cdot \begin{bmatrix} v_1 \\ v_2 \\ v_3 \\ \vdots \\ v_n \end{bmatrix} = u_1v_1 + u_2v_2 + u_3v_3 + \cdots + u_nv_n$$

This product takes two vectors in R^n to a single number in R .We will focus in on this very important product in chapter 8 when we talk about orthogonality. As it turns out when the dot product of two vectors is zero they are at 90° to eachother. It cal also be used to determine how much of one vector is in the direction of the other.

As we will find out the dot product is an example of mapping from a space $S \to R$ called an inner product. For now we will focus on the dot product itself, an inner product taking $R^n \to R$.

Let's get some properties of the dot product out of the way,

Let $\vec{u}, \vec{v}, \vec{w} \in R^n$ and $c \in R$ then

(a) $\vec{u} \cdot \vec{v} = \vec{v} \cdot \vec{u}$
(b) $(\vec{u} + \vec{v}) \cdot \vec{w} = (\vec{u} \cdot \vec{w} + \vec{v} \cdot \vec{w})$
(c) $(c\vec{u}) \cdot \vec{v} = c(\vec{u} \cdot \vec{v}) = \vec{u} \cdot (c\vec{v})$
(d) $\vec{u} \cdot \vec{u} \geq 0, and\ \vec{u} \cdot \vec{u} = 0\ if\ and\ only\ if\ \vec{u} = \vec{0}$

Proof of (a)

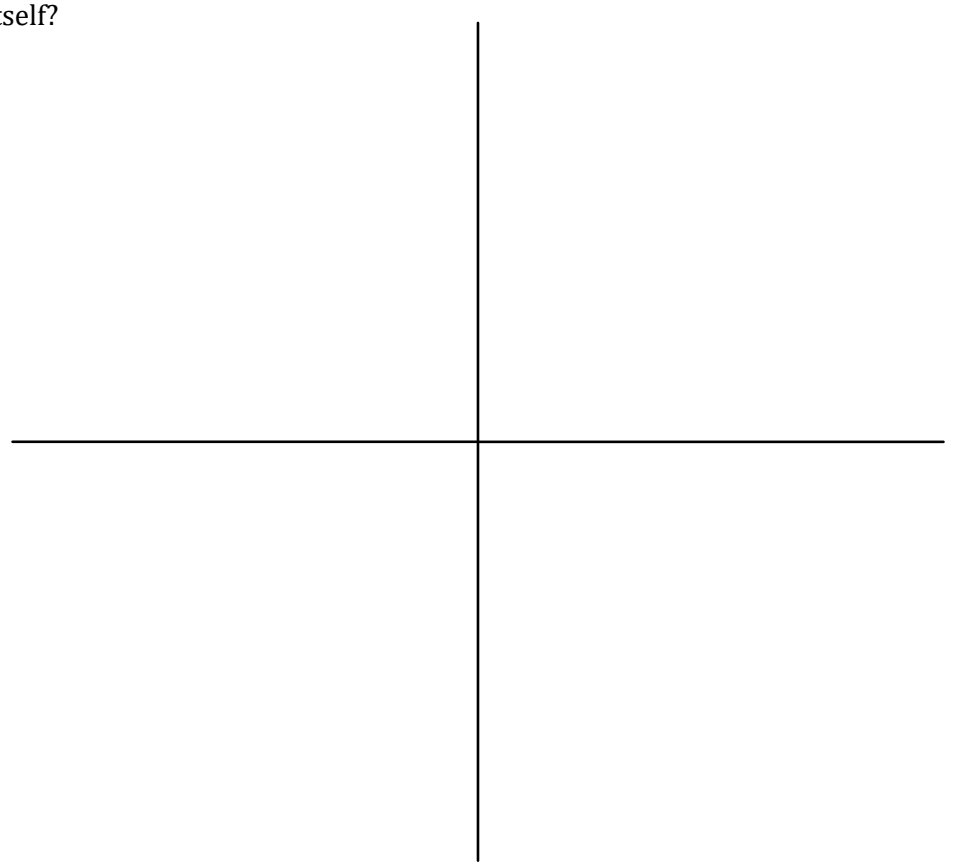
What about $\vec{u} \cdot (\vec{v} \cdot \vec{w}) = (\vec{u} \cdot \vec{v}) \cdot \vec{w}$?

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Dot product to measure distance

What happens when we dot a vector with itself?

Example: calculate $\vec{u} \cdot \vec{u} = \begin{bmatrix} 2 \\ 1 \end{bmatrix} \cdot \begin{bmatrix} 2 \\ 1 \end{bmatrix}$



You might recall from sec 2.1 that the length of a vector is given by:

Magnitude of a vector let $\vec{u} \in \mathbf{R^n}$ then the magnitude is $|\vec{u}| = \sqrt{u_1^2 + u_2^2 + u_3^2 \cdots u_n^2}$

In fact:

$$|\vec{u}| = \sqrt{\vec{u} \cdot \vec{u}}$$

We call this the norm of $\vec{u}$

Let $\vec{u} \in R^n$ then the norm(or length of) $\vec{u}$ is given by

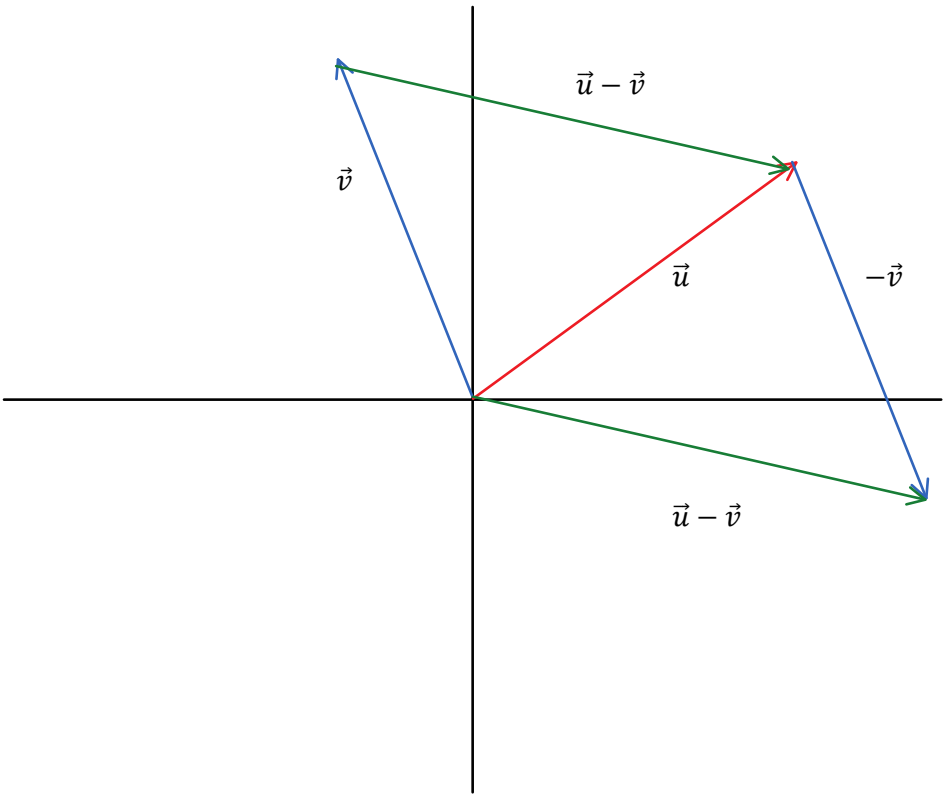
$$\left||\vec{u}|\right| = \sqrt{\vec{u} \cdot \vec{u}}$$

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Distance between two vectors

Let $\vec{u}, \vec{v} \in R^n$ then the distance between $\vec{u}$ *and* $\vec{v}$ is given by

$$d(\vec{u}, \vec{v}) = ||\vec{u} - \vec{v}||$$



The dot product can also tell us the angle between two vectors.

Let $\vec{u}, \vec{v} \in R^n$ then

$$\vec{u} \cdot \vec{v} = ||u||||v||\cos(\theta)$$

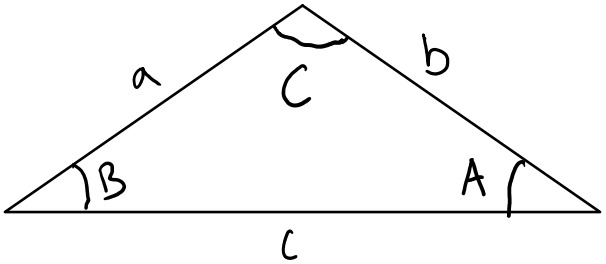
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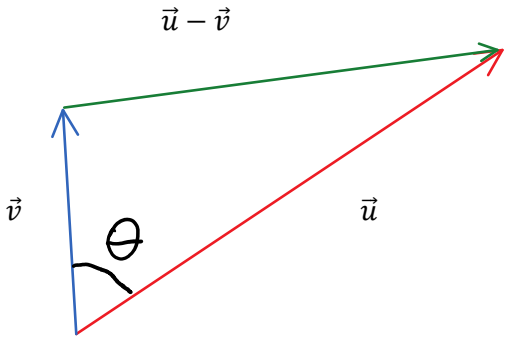
Let $\vec{u}, \vec{v} \in R^n$ then

$$\vec{u} \cdot \vec{v} = ||u||||v||\cos(\theta)$$

Proof: Let $\vec{u}, \vec{v} \in R^n$

Think about the fact that they form a triangle in a plane.





Law of Cosines: $c^2 = a^2 + b^2 - 2abcos(C)$

Law of Cosines: $\left|\left|\vec{u} - \vec{v}\right|\right|^2 = \left|\left|\vec{u}\right|\right|^2 + \left|\left|\vec{v}\right|\right|^2 - 2\left|\left|\vec{u}\right|\right|\left|\left|\vec{v}\right|\right|\cos(\theta)$

$$\left|\left|\vec{u} - \vec{v}\right|\right|^2 = \left|\left|\vec{u}\right|\right|^2 - \left|\left|\vec{v}\right|\right|^2 + 2\left|\left|\vec{u}\right|\right|\left|\left|\vec{v}\right|\right|\cos(\theta)$$

$$(\vec{u} - \vec{v}) \cdot (\vec{u} - \vec{v}) = \vec{u} \cdot \vec{u} - \vec{v} \cdot \vec{v} + 2(\vec{u} \cdot \vec{u})(\vec{v} \cdot \vec{v})\cos(\theta)$$

$$\vec{u} \cdot \vec{u} - (\vec{u} \cdot \vec{v}) - (\vec{v} \cdot \vec{u}) + \vec{v} \cdot \vec{v} - = \vec{u} \cdot \vec{u} + \vec{v} \cdot \vec{v} - 2(\vec{u} \cdot \vec{u})(\vec{v} \cdot \vec{v})\cos(\theta)$$

$$-(\vec{u} \cdot \vec{v}) - (\vec{v} \cdot \vec{u}) = -2(\vec{u} \cdot \vec{u})(\vec{v} \cdot \vec{v})\cos(\theta)$$

$$-2(\vec{u} \cdot \vec{v}) = -2(\vec{u} \cdot \vec{u})(\vec{v} \cdot \vec{v})\cos(\theta)$$

$$(\vec{u} \cdot \vec{v}) = (\vec{u} \cdot \vec{u})(\vec{v} \cdot \vec{v})\cos(\theta)$$

$$\vec{u} \cdot \vec{v} = ||u||||v||\cos(\theta)$$

So what happens when the angle between them is 90° ?

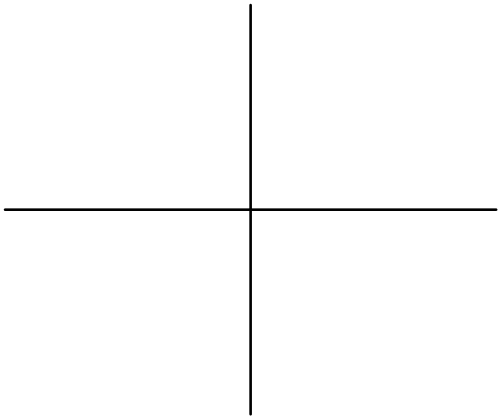
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Vectors $\vec{u}, \vec{v} \in R^n$ are orthogonal if and only if

$$\vec{u} \cdot \vec{v} = 0$$

Pythagorean theorem

$$Let\ \vec{u}, \vec{v} \in R^n\ then\ \left|\left|\vec{u} + \vec{v}\right|\right|^2 = \left|\left|\vec{u}\right|\right|^2 + ||v||^2\ if\ and\ only\ if\ \vec{u} \cdot \vec{v} = 0$$



$$Cauchy - Schwarz\ inequality: if\ \vec{u}, \vec{v} \in R^n\ then$$

$$\left|\vec{u} \cdot \vec{v}\right| \leq ||\vec{u}||||\vec{v}||$$

Proving one of these would make a great quiz problem!

The proofs are easily found online.

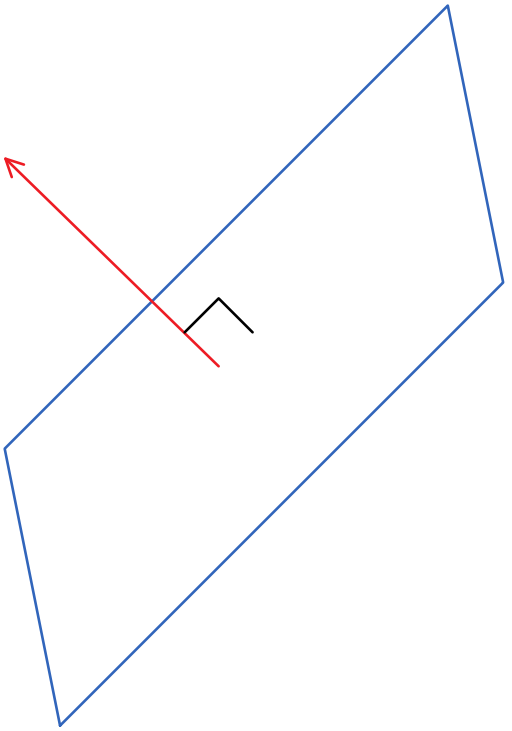
Triangle Inequality: $if\ \vec{u}, \vec{v} \in R^n\ then$

$$\left|\left|\vec{u} + \vec{v}\right|\right| \leq \left|\left|\vec{u}\right|\right| + \left|\left|\vec{v}\right|\right|$$

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Let S be a subspace of R^n. We say that a vector $\vec{u} \in \mathrm{R}^n$ is orthognal to S if $\vec{u} \cdot \vec{s} = 0$ for every $\vec{s} \in \mathrm{S}$. The set of all such vectors is called the orthognal complement of S and is denoted $S^{\perp}$

An easy example to visualize is that of a vector which is perpendicular to a plane. In fact that vector can define the plane.



Let $\vec{u} = \begin{bmatrix} a \\ b \\ c \end{bmatrix} \in R^3$ then the equation of the plane permindicular to $\vec{u}$ going containing a point (x_0, y_0, z_0) is given by:

$$a(x - x_0) + b(y - y_0) + c(z - z_0) = 0$$

Which is the same as the requirement

$\begin{bmatrix} a \\ b \\ c \end{bmatrix} \cdot \begin{bmatrix} x - x_0 \\ y - y_0 \\ z - z_0 \end{bmatrix} = 0$ that is, that $\vec{u}$ is perpindicular to every vector in the plane.

If S is a subspace of R^n then so is $S^{\perp}$.

Proof:

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Let $V = \{\vec{s}_1, \ldots, \vec{s}_k\}$ be a basis for a subspace S and let $\vec{u}$ be a vector. Then $\vec{u} \in S^{\perp}$ if and only if

$$\vec{u} \cdot \vec{s}_1 = \vec{u} \cdot \vec{s}_2 = \cdots = \vec{u} \cdot \vec{s}_k = 0$$

Proof: Suppose $\vec{u} \in S^{\perp}$, then by definition it is orthognal to all vectors in S and thus orthognal to the basis vectors.

Now suppose $\vec{u} \cdot \vec{s}_1 = \vec{u} \cdot \vec{s}_2 = \cdots = \vec{u} \cdot \vec{s}_k = 0$ and let $\vec{s} \in S$ then because V is a basis we can say,

$$\vec{s} = c_1\vec{s}_1 + c_2\vec{s}_2 + \cdots + c_k\vec{s}_k$$

And then

$$\vec{u} \cdot \vec{s} = \vec{u} \cdot (c_1\vec{s}_1 + c_2\vec{s}_2 + \cdots + c_k\vec{s}_k) = \vec{u} \cdot c_1\vec{s}_1 + \vec{u} \cdot c_2\vec{s}_2 + \cdots + \vec{u} \cdot c_k\vec{s}_k = c_1\vec{u} \cdot \vec{s}_1 + c_2\vec{u} \cdot \vec{s}_2 + \cdots + c_k\vec{u} \cdot \vec{s}_k$$

$$= c_1(0) + c_2(0) + \cdots + c_k(0) = 0$$

So for every $\vec{s} \in S$ $\vec{u} \cdot \vec{s} = 0$ thus $\vec{u} \in S^{\perp}$.

A set of vectors V in R^n form an orthognal set if and only if $\vec{v}_i \cdot \vec{v}_j = 0$ whenever $i \neq j$ for all vectors in V. Another way to state this is $\vec{v}_i \cdot \vec{v}_j = \delta_{ij}$.

An orthogonal set of vectors $V = \{\vec{v}_1 \ldots \vec{v}_k\}$ is linearly independent

Proof:

Let $\{\vec{v}_1 \ldots \vec{v}_k\}$ be an orthognal set of vectors.

$$c_1\vec{v}_1 + \cdots + c_k\vec{v}_k = \vec{0}$$

$$\vec{v}_1 \cdot (c_1\vec{v}_1 + \cdots + c_k\vec{v}_k) = \vec{v}_1 \cdot \vec{0}$$

$$c_1\vec{v}_1 \cdot \vec{v}_1 + c_2\vec{v}_1 \cdot \vec{v}_2 + \cdots + c_k\vec{v}_1 \cdot \vec{v}_k = \vec{0}$$

Consider

$$c_1\left|\left|\vec{v}_1\right|\right|^2 + c_2(0) + \cdots c_k(0) = \vec{0}$$

So since $\vec{v}_1 \neq \vec{0}$, $\left|\left|\vec{v}_1\right|\right|^2 \neq 0$ thus $c_1 = 0$. Repeat with all of the vectors in $V$to find that indeed if $c_1\vec{v}_1 + \cdots + c_k\vec{v}_k = \vec{0}$ then it must be that $c_1 = c_2 = \cdots = c_k = 0$ the the vectors are linearly independent.

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Let $V = \{\vec{v}_1 \ldots \vec{v}_k\}$ be an orthognal basis for a subspace S, then for any vector $\vec{s} \in S$ we can write $\vec{s}$ as

$$\vec{s} = c_1\vec{v}_1 + c_2\vec{v}_2 + \cdots + c_k\vec{v}_k \;\; \textit{where } c_i = \frac{\vec{v}\cdot\vec{s}}{||\vec{v}_i||}$$

Proof: Again consider $\vec{v}_1 \cdot \vec{s} = \vec{v}_1 \cdot \left(c_1\vec{v}_1 + \cdots + c_k\vec{v}_k\right)$ and let's see if we can solve for c_1.

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Find a basis for $S^{\perp}$ for the subspace S.

$$S = \text{span}\left\{\begin{bmatrix} 1 \\ 1 \\ -7 \end{bmatrix}\right\}$$

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$$\left\{ \begin{bmatrix} \square \\ \square \\ \square \end{bmatrix} \right\}$$

[-1, 7; 1, 0; 0, 1]